Using **mmWave** to **Simulate** 5G NETWORKS

5G is the fifth generation of mobile cellular technology. There are both high expectations from and fears about this technology. Generally, it is expected that communication will be speedier with 5G technology, with speeds of up to 10Gbps using millimetre waves of 15GHz.

These days, most government and corporate services are available on mobile networks, and smartphones are connected with high performance 4G or 3G cellular networks. The existing technologies for smartphones as well as traditional mobile networks have evolved to the extent that mobility doesn't affect performance, and users access the services with minimum delays and a high degree of speed. The current technologies and protocols of mobile networks are required for high efficiency as most of the services including e-governance applications, higher education services, corporate applications, stock market predictions, e-commerce, conferencing based communications, business communications, e-learning based applications, etc, are based on real-time communications.

As there are different technologies and delivery models for services offered by ISPs, there is a need to be able to predict network performance in different environments so that the behaviour of a new mobile network technology can be controlled before actual deployment in a real-time customer-facing environment.

If the new cellular network technology is deployed without simulation, it can be very dangerous for the users as well as the service providers.

Figure 1 depicts the huge growth in the usage patterns of mobile Internet up to 2022 and shows the scope for improvements in the existing cellular technologies in order to achieve a higher degree of accuracy and performance.

There are a number of tools and libraries for the prediction of mobile network behaviour, which can be used to generate virtual nodes, towers, base stations, controllers and servers. Table 1 lists prominent tools and frameworks used for network simulation. These tools and libraries can be used to test the behaviour of a new network without actually having the smartphones, devices, computers, towers and other related physical infrastructure in place.